# Leandro Sánchez-Betancourt

*Curriculum vitae*

*S1.30 Mathematical Institute*
*University of Oxford*
*OX2 6GG, Oxford, United Kingdom*
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## Academic posts

2025–present **Associate Professorship**, *Mathematical Institute, University of Oxford and Oxford-Man Institute of Quantitative Finance*.

2023–present **Senior Research Fellowship**, *Mathematical Institute, University of Oxford and Oxford-Man Institute of Quantitative Finance*, Mathematical and Computational Finance research group.

2023–2025 **Non-Stipendiary Lectureship in Mathematics**, *New College, University of Oxford*, Probability and Statistics.

2022–2023 **Lectureship (Assistant Professorship)**, *Department of Mathematics, King's College London*, Financial Mathematics research group.

2021 **Research Associateship**, *Department of Mathematics, Imperial College London*, Mathematical Finance Section.

2019–2021 **Graduate Teaching and Research Scholarship**, *Oriel College, Oxford*.

## Education

2017–2021 **DPhil in Mathematics**
**University of Oxford**, Mathematical Institute.
Dissertation: Uncertain Execution in Order-driven Markets.
Supervisor: Professor Álvaro Cartea.
Prize: Bruti Liberati prize for best PhD thesis in Quantitative Finance.

2016–2017 **MSc in Financial Mathematics**
**King's College London**, Department of Mathematics.
Overall mark: Distinction (94/100).
Prize: Best overall performance student award.

2010–2014 **BSc in Actuarial Sciences**
**Universidad Nacional Autónoma de México**, Facultad de Ciencias.
Overall mark: 10/10.
Prize: Gabino Barreda Medal (for top student).

## Awards, accomplishments, and scholarships

2025 Award of title of Associate Professor. Mathematical Institute, University of Oxford.

2023 Bruti Liberati Prize for best PhD thesis in Quantitative Finance €3,000.
Awarded by the Bachelier Finance Society and the Politecnico di Milano, in cooperation with Springer.
Committee: Prof. E. Barucci, Prof. J. Cvitanić, Prof. M. Grasselli, Prof. X. Guo, Prof. H. Xing

2021–2023 Co-investigator in the AFM-ATI collaboration (Authority for the Financial Markets, Netherlands and Alan Turing Institute, London). Funding for the project £249,000.

2019–2021 Graduate Teaching and Research Scholarship in Mathematics, Oriel College, Oxford ∼ £20,000.

2019 Winner of the Financial Mathematics Team Challenge (as Team Leader), Rio de Janeiro, Brazil.

2018–2021 Scholarship for doctoral studies at Oxford, awarded by Consejo Nacional de Ciencia y Tecnología (CONACyT), México ∼ £50,000.

2017–2020 Mathematical Finance Scholarship, awarded by Mathematical Institute, Oxford £24,000.

2017 Best overall performance student award, for the highest marks in the MSc in Financial Mathematics, Department of Mathematics, King's College London.

2016 Scholarship for graduate studies at King's College London, awarded by CONACyT, México ∼ £22,000.

2016 Scholarship for graduate studies at King's College London, awarded by Secretaría de Educación Pública (SEP), México ∼ £3,000.

2016 Gabino Barreda medal, for top student in four-year undergraduate program, awarded by Universidad Nacional Autónoma de México (UNAM).

2011,-12,-13 Excellence Scholarship, awarded by SEP.

2008,-09 Winner of the Mathematical Olympiad of México, at state level.

2007,-08,-09 Winner of the Mathematical Competition, over all the schools (UNAM).

2004,-05 Winner of the Mathematical Olympiad of Cuba, at national level.

## Membership of professional societies

Society for Industrial and Applied Mathematics (SIAM)
Bachelier Finance Society (BFS)
London Mathematical Society (LMS)
Institute of Mathematics and its Applications (IMA)

## Journal articles

[23] Á. Cartea, F. Drissi, L. Sánchez-Betancourt, D. Siska, L. Szpruch (2026). *Strategic Bonding Curves in Automated Market Makers*.
**Mathematics of Operations Research**.

[22] E. Barucci, A. Mathieu, L. Sánchez-Betancourt (2026). *Market Making with Fads, Informed, and Uninformed Traders*.
**Mathematical Finance**.

[21] S. N. Cohen, L. Sánchez-Betancourt, L. Szpruch (2026). *Interest Rate Models in Decentralised Lending Protocols*.
**Digital Finance**.

[20] A. Aqsha, P. Bergault, L. Sánchez-Betancourt (2026). *Equilibrium Reward for Liquidity Providers in Automated Market Makers*.
**Mathematical Finance**.

[19] Á. Cartea, S. Jaimungal, L. Sánchez-Betancourt (2026). *Nash Equilibrium between Brokers and Traders*.
**Finance and Stochastics** **30** (3), 951–982.

[18] Á. Cartea, G. Duran-Martin, L. Sánchez-Betancourt (2026). *Detecting Toxic Flow*.
**Quantitative Finance**, **26** (4), 541-561.

[17] T. Bhudisaksang, Á. Cartea, L. Sánchez-Betancourt (2026). *Adaptive-Robust Portfolio Optimisation*.
**Mathematics and Financial Economics** **20** (1), 171-202.

[16] R. Boyce, M. Herdegen, L. Sánchez-Betancourt (2025). *Market Making with Exogenous Competition*.
**SIAM Journal on Financial Mathematics** **16** (2), 692-706.

[15] G. Duran-Martin, L. Sánchez-Betancourt, A. Shestopaloff, K. Murphy (2025). *A unifying framework for generalised Bayesian online learning in non-stationary environments*.
**Transactions on Machine Learning Research**.

[14] P. Bergault, L. Sánchez-Betancourt (2025). *A Mean Field Game between Informed Traders and a Broker*.
**SIAM Journal on Financial Mathematics** **16** (2), 358-388.

[13] Á. Cartea, L. Sánchez-Betancourt (2025). *Brokers and Informed Traders: dealing with toxic flow and extracting trading signals*.
**SIAM Journal on Financial Mathematics** **16** (2), 243-270.

[12] Á. Cartea, S. N. Cohen, R. Graumans, S. Labyad, L. Sánchez-Betancourt, L. van Veldhuijzen (2025). *Statistical Predictions of Trading Strategies in Electronic Markets*.
**Journal of Financial Econometrics** **23** (2), nbae025.

[11] L. P. Hughston, L. Sánchez-Betancourt (2024). *Valuation of a Financial Claim Contingent on the Outcome of a Quantum Measurement*.
**Journal of Physics A: Mathematical and Theoretical** **57** (28), 285302.

[10] G. Bouzianis, L. P. Hughston, L. Sánchez-Betancourt (2024). *Information-based Trading*.
**International Journal on Theoretical and Applied Finance** **27** (March, April), 2350030.

[9] S. Jaimungal, S. Pesenti, L. Sánchez-Betancourt (2024). *Minimal Kullback-Leibler for Constrained Lévy-Itô Processes*.
**SIAM Journal on Control and Optimization 60** (2), 982-1005.

[8] C. Bellani, D. Brigo, M. S. Pakkanen, L. Sánchez-Betancourt (2024). *Price impact without averaging*.
**Applied Mathematical Finance 30** (4), 175-206.

[7] Á. Cartea, L. Sánchez-Betancourt (2023). *Optimal Execution with Stochastic Delay*.
**Finance and Stochastics 27** (1), 1-47

[6] Á. Cartea, I. Perez Arribas, L. Sánchez-Betancourt (2022). *Double-Execution Strategies using Path Signatures*.
**SIAM Journal on Financial Mathematics 13** (4), 1379–1417.

[5] M. Forde, L. Sánchez-Betancourt, B. Smith (2022). *Optimal Trade Execution for Gaussian Signals with Power-law Resilience*.
**Quantitative Finance 22** (3), 585–596.

[4] Á. Cartea, S. Jaimungal, L. Sánchez-Betancourt (2021) *Latency and Liquidity Risk*.
**International Journal on Theoretical and Applied Finance 24** (June, July), 2150035.

[3] G. Bouzianis, L. P. Hughston, S. Jaimungal, L. Sánchez-Betancourt (2021). *Lévy-Ito Models in Finance*.
**Probability Surveys 18**, 132-178.

[2] Á. Cartea, L. Sánchez-Betancourt (2021). *The Shadow Price of Latency: Improving Intraday Fill Ratios in Foreign Exchange Markets*.
**SIAM Journal on Financial Mathematics 12** (1), 254–294.

[1] L. P. Hughston, L. Sánchez-Betancourt (2020). *Pricing with Variance Gamma Information*.
**Risks 8** (4), 105:1-22.

## Conference papers, workshop papers and book chapters

[9] H. Yiu, L. Sánchez-Betancourt, Á. Cartea, G. Duran-Martin (2026). *Doubly Outlier-Robust Online Infinite Hidden Markov Model*.
**International Conference on Machine Learning (ICML)**.

[8] K. Li, J. Miao, M. Cucuringu, L. Sánchez-Betancourt (2025). *LLM Embedding for Regression Priors*.
**6th ACM International Conference on AI in Finance (ICAIF)**.

[7] G. Duran Martin, L. Sánchez-Betancourt, Á. Cartea, K. P. Murphy (2025). *Martingale Posterior Neural Networks for Fast Sequential Decision Making*.
**Conference on Neural Information Processing Systems (NeurIPS)**.

[6] K. Li, M. Cucuringu, L. Sánchez-Betancourt, T. Willi (2024). *Mixtures of Experts for Scaling up Neural Networks in Order Execution*.
**5th ACM Internaltional Conference on AI in Finance (ICAIF)**.

[5] A. Bogdan, L. Sánchez-Betancourt, S. Sarkadi, C. Ventre (2024). *Detecting Collective Liquidity Taking Distributions*.
**5th ACM Internaltional Conference on AI in Finance (ICAIF)**.

[4] G. Duran-Martin, M. Altamirano, A. Shestopaloff, L. Sánchez-Betancourt, J. Knoblauch, M. Jones, F-X. Briol, K. P. Murphy (2024). *Outlier-robust Kalman Filtering through Generalised Bayes*.
**International Conference on Machine Learning (ICML)**.

[3] J. Jerome, L. Sánchez-Betancourt, R. Savani, M. Herdegen (2023). *Model-based gym environments for limit order book trading*.
**4th ACM International Conference on AI in Finance (ICAIF)**.

[2] M. Höglund, E. Ferrucci, C. Hernández, A. Muguruza Gonzalez, C. Salvi, L. Sánchez-Betancourt, Y. Zhang (2023). *A Neural RDE approach for continuous-time non-Markovian stochastic control problems*.
*Workshop on New Frontiers in Learning, Control, and Dynamical Systems* at the **International Conference on Machine Learning (ICML)**.

[1] Á. Cartea, S. Jaimungal, L. Sánchez-Betancourt (2023). *Reinforcement Learning for Algorithmic Trading*.
**Machine Learning and Data Sciences for Financial Markets.** Edited by C.-A. Lehalle and A. Capponni. Cambridge University Press.

## Submitted papers* and working papers+

+ D. Itkin, L. Sánchez-Betancourt (2026). *When Hedging Changes the Payoff: Option Replication with Price Impact and Execution Costs*.
* L. Baggiani, M. Herdegen, L. Sánchez-Betancourt (2026). *Competition between DEXs through Dynamic Fees*.
+ A. Aqsha, P. Bank, L. Sánchez-Betancourt (2026). *Solving Linear-Quadratic Stochastic Control Problems with Signatures*.
+ N. Chilenje, M. Daba, D. Feleppa, C. Fellner, L. Sánchez-Betancourt (2025). *Market Making with Competition*.
* P. Bergault, S. Bieber, L. Sánchez-Betancourt (2025). *Optimal Exit Time for Liquidity Providers in Automated Market Makers*.
* L. Baggiani, M. Herdegen, L. Sánchez-Betancourt (2025). *Optimal Dynamic Fees in Automated Market Makers*.
* Á. Cartea, L. Sánchez-Betancourt (2025). *A Simple Strategy to Deal with Toxic Flow*.
+ A. Aqsha, F. Drissi, L. Sánchez-Betancourt (2024). *Strategic Learning and Trading in Broker-Mediated Markets*.

## Industry experience

2017–2024 **Researcher**, *LMAX Exchange*, London.
Investigated topics regarding latency and high-frequency trading.
Made comparisons between trading-on-firm venues and last-look venues.
Developed new frameworks for solving externalisation-internalisation problems.

2015–2016 **Consultant**, *Indra Business Consulting*, Mexico City.
Provided mathematical support for the migration of Santander's risk-management platform.
Undertook mathematical analysis of the methodological changes in derivative pricing in Murex.

2014–2015 **Risk analyst**, *Citigroup*, Mexico City.
Developed internal debt rating model in collaboration with Citigroup New York.
Carried out monthly calculations of the regulatory reserves.

## Teaching

2024–2026 **Lecture course**, *Mathematical Institute, University of Oxford*.
Financial Derivatives (Michaelmas term), Market Microstructure and Algorithmic Trading (Hilary term), Decentralised Finance (Hilary term).

2026 **Lecture course**, *Saïd Business School, University of Oxford*.
Continuous Time Finance (Trinity term).

2024–2026 **Reading groups**, *Oxford-Man Institute of Quantitative Finance, University of Oxford*.
Lectures on BSDEs, Stochastic Control, and Stochastic Differential Games with Financial Applications (Michaelmas term 2024). Market Microstructure (Michaelmas term 2025, Hilary term 2026, Michaelmas term 2027).

2024–2026 **Lecture course**, *Oxford-Man Institute of Quantitative Finance, University of Oxford*.
Machine Learning in Finance.

2023–2025 **Tutorials**, *New College, Oxford*.
Probability, Statistics

2023–2024 **Lecture course**, *Mathematical Institute, University of Oxford*.
Fixed Income and Credit

2022–2023 **Lecture course**, *King's College London*.
Mathematical Finance II: Continuous Time

2021 **Lecture course**, *Imperial College London*.
Quantitative Risk Management (Core module for the MSc Mathematics and Finance)

2019–2021 **Tutorials**, *Oriel College, Oxford*.
Probability, Probability prelims (Michaelmas term 2019, 2020), Integration, Statistics, Integral transforms (Hillary term 2020, 2021), Statistics and data analysis (Trinity term 2020, 2021).

2018–2019 **Tutorials**, *Queen's College, Oxford*.
Probability (Michaelmas term 2018), Statistics and data analysis (Trinity term 2019).

2018–2021 **Tutorials**, *Mathematical Institute, University of Oxford*.
Graduate courses: Algorithmic trading, Asset pricing, Stochastic control, Market micro-structure.
Undergraduate courses: Stochastic differential equations, Mathematical models for financial derivatives.

## Presentations at conferences, seminars and summer schools

(*invited talks)

2026 AIFMRM seminar, University of Cape Town.*
SIAM annual meeting, Cleveland, Ohio.
13th World Congress of the Bachelier Finance Society, Bologna.
Financial Engineering seminar, Chinese University of Hong Kong.
Oxford-PKU Joint Conference on Quantitative Finance and Data Science, Peking University.*
Financial Mathematics seminar, Politecnico di Milano.*
Computational Economics and Finance Seminar Series, University College London.*
Quantitative Finance seminar, University of Birmingham.*

2025 Financial Mathematics seminar, London School of Economics.*
Finance and Stochastics seminar, Imperial College London.*
Advances in Mathematics of Randomness for Handling Risks in Finance and Insurance, Centre International de Rencontres Mathématiques (CIRM), Luminy.*
SIAM Conference on Financial Mathematics, Miami.
Financial Mathematics seminar at University of Cape Town, South Africa.*
Advances in Stochastic Control and Reinforcement Learning Workshop, Banff, Canada.*

2024 Algo-Trading & DeFi Workshop, Politecnico di Milano, Milan.*
Stochastic Finance at Warwick seminar, Coventry.*
Mathematics of Random Systems workshop, Oxford.*
Workshop between the Authority for the Financial Markets (AFM) and the Financial Conduct Authority (FCA), London.*
Oxford-Princeton Conference on Mathematical Finance, Princeton University, New Jersey.
Manchester Probability Seminar, University of Manchester.*
Institute for Mathematical Sciences (IMS) Young Mathematical Scientist Forum in Applied Mathematics, Singapore.*

2023 4th Association for Computing Machinery (ACM) International Conference on AI in Finance, New York.
Berlin Seminar on Stochastic Analysis and Stochastic Finance, Humboldt University of Berlin.*
Mathematical Finance Internal Seminar, University of Oxford.*
Financial and Actuarial Mathematics at UCLA, Los Angeles.*
Financial Engineering seminar, Politecnico di Milano.*
Mathematical finance seminar at Columbia, New York.*
Blockchain@X-OMI Workshop on Blockchain and Decentralized Finance, Paris.*
European Finance Association annual meeting, Amsterdam.*
11th Advanced Mathematical Methods for Finance (AMaMeF) Conference, Bielefeld.*
SIAM Conference on Financial Mathematics, Philadelphia.
Man Group research seminar, London.*
Financial Technology conference, Oxford-Man Institute of Quantitative Finance, Oxford.*
University of Edinburgh Quantitative Finance seminar, Edinburgh.*

2022 Quantitative Finance Seminar, Fields Institute, Toronto.*
Finance and Economics, Alan Turing Institute, London.*
Stochastic Finance at Warwick seminar, Coventry.*
SIAM Conference on Mathematics of Data Science, San Diego.
11th World Congress of Bachelier Finance Society, Hong Kong.
Capital Fund Management (CFM) EconophysiX Lab seminar, Paris.*
University of Edinburgh Quantitative Finance seminar, Edinburgh.*
1st London/Oxford/Warwick Financial Mathematics Workshop, London.*

2021 Conference on Computational and Financial Econometrics (CFE) 2021, King's College London.*
Big Data and Machine Learning in Finance Conference, Politecnico di Milano.
SIAM Conference on Financial Mathematics, virtual.
Industrial Engineering and Operations Research (IEOR) seminar, University of California at Berkeley.

2020 Mathematical Finance Internal Seminar, University of Oxford.

2019 21st Actuarial Congress: "un modelo a seguir", Universidad Marista, Mexico City.*
12th European Summer School on Financial Mathematics, Padova.
SIAM Conference on Financial Mathematics, Toronto.
1st Oxford-ETH Workshop on Financial Mathematics, Oxford.

2018 11th European Summer School on Financial Mathematics, Paris.
10th World Congress of the Bachelier Finance Society, Dublin.
Actuarial Sciences Conference, Universidad Marista, Mexico City.
Mathematical Finance Internal Seminar, University of Oxford.
Market Microstructure, Imperial-CFM workshop, London.

## Service to the community

2024–present Associate Editor of São Paulo Journal of Mathematical Sciences, Springer.

Refereeing SIAM Journal on Financial Mathematics (SIFIN). Operations Research (OR). Finance and Stochastics (Finance Stoch). Mathematical Finance (MAFI). Quantitative Finance (QF). Applied Mathematical Finance (AMF). International Conference on AI in Finance (ICAIF). AAAI workshops. Probability, Uncertainty and Quantitative Risk (PUQR). Economic Modelling (EM). International Journal of Theoretical and Applied Finance (IJTAF). Risk Cutting Edge (Risk.net). Journal of Financial Econometrics (JFEC). Applied Mathematics and Optimization (AMOP). Mathematics and Financial Economics (MAFE). Finance Research Letters (Fin Res Lett). Dynamic Games and Applications (Dyn Games Appl).

Conference organisation Scientific and Organising Committee: 3nd Algo-Trading & DeFi Workshop 2026.
Scientific and Organising Committee: Oxford-Princeton Conference 2025.
Scientific and Organising Committee: 2nd Algo-Trading & DeFi Workshop 2025.
Mini-Symposium Organiser at: SIAM Conference on Financial Mathematics and Engineering 2021, 2023, 2025, and Bachelier world congress 2024.
Co-organiser of the Seminar Series on "Statistics and Machine Learning in Finance".
Workshop Co-Chair for 2023 International Conference on AI in Finance (ICAIF).
Scientific and Organising Committee: London-Oxford-Warwick Mathematical Finance Workshop 2024.
Scientific and Organising Committee: OMI Statistical Machine Learning and Finance Workshop 2024.

## Examiner for transfer of status ($*$), confirmation ($\star$), and PhD viva ($+$)

2026 Valentin Mohl* (Oxford), Yichi Zhang* (Oxford), Yanzhao Yang* (Oxford), Danni Shi* (Oxford), Yuantao Shi$^{+}$ (Oxford), Ruotong Cao* (Oxford).

2025 Jakob Albers* (Oxford), Michael Giegrich$^{+}$ (Oxford), Shijia Jin$^{+}$ (Monash University), Patrick Chang$^{+}$ (Oxford), Jason Rader* (Oxford), Gabriel Garcia Arenas$^{+}$ (Oxford), Zi Li$^{+}$ (King's College London), Nicolas Petit* (Oxford), Guangyi He (Imperial College London).

2024 Gabriel Garcia Arenas* (Oxford), Jakob Albers* (Oxford), Kang Li* (Oxford), Yanzhao Yang* (Oxford), Laura Körber$^{+}$ (TU Berlin), Marcello Monga$^{+}$ (Oxford), Jason Rader* (Oxford), Cephas Svosve* (Oxford), Zihan Guo* (Oxford).

2023 Karolina Bassa* (Oxford).

## Academic visits, outreach, and team projects

2026–present **Research collaboration with Banca Sella, Piedmont, Italy.**
Investigated competition between market makers with proprietary data.

Sept 2026 **Academic visit to Université Paris-Dauphine, France**.
Collaborated with Professor Philippe Bergault.

Mar 2026 **Outreach Target Oxbridge, Oxford, England.**
Lectured on "Large(ish) Language Models".

Jun 2025 **Financial Mathematics Team Challenge, Cape Town, South Africa**.
My role was that of a mentor. The team that I mentored won the competition.

Dec 2024 **Academic visit to Université Paris-Dauphine, France**.
Collaborated with Professor Philippe Bergault.

Oct 2024 **London Mathematical Society, Research in Pairs, University of Oxford**.
Visit from Professor Lane P. Hughston to carry out joint research under an LMS Research in Pairs grant.

Sept 2022 **Autoriteit Financiële Markten, Amsterdam, The Netherlands**.
Researched agent-based market simulators, clustering of trading algorithms, and detection of market manipulation.

Jul 2019 **Financial Mathematics Team Challenge, Rio de Janeiro, Brazil**.
My role was that of team leader. Our team won the competition.

Jun 2019 **Fields-China Joint Industrial Problem Solving Workshop in Finance, Fields Institute, Toronto**.
Deep Machine Learning and Volatility Prediction.

Jun 2018 **Academic visit to University of Toronto, Canada.**
Collaborated with Professor Sebastian Jaimungal.

2018–2020 **Christopher Hatton School, London**.
Led a weekly Math Club. Coached the students to solve Mathematical Olympiad problems.

## References


1. **Professor Álvaro Cartea**
Mathematical Institute, University of Oxford, OX2 6GG, Oxfordshire, United Kingdom.
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2. **Professor Samuel N. Cohen**
Mathematical Institute, University of Oxford, OX2 6GG, Oxfordshire, United Kingdom.
*samuel.cohen@maths.ox.ac.uk*
3. **Professor Mihai Cucuringu**
Department of Mathematics, UCLA, MS 6363, Los Angeles, United States of America.
*mihai@math.ucla.edu*
4. **Professor Lane P. Hughston**
School of Computing, Goldsmiths University of London, SE14 6NW, London, United Kingdom.
*l.hughston@gold.ac.uk*
5. **Professor Sebastian Jaimungal**
Department of Statistical Sciences, University of Toronto, M5T 1P5, Ontario, Canada.
*sebastian.jaimungal@utoronto.ca*